Reproducible Science: Machine-Checkable Claims from Preregistration to Publication

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Abstract

Every claim in a paper comes from somewhere. AI tooling allows us to quickly generate hypotheses, run analyses, and draft manuscripts, but we identify verification and provenance as open issues: (1) verifying that the finished paper agrees with the artifacts behind it typically requires manual or agentic verification, which is expensive and difficult to audit; (2) verification usually does not include a denominator for the number of sources or artifacts checked; (3) verification snapshots quickly become stale, and are not guaranteed to find every issue or reproduce the same issues on subsequent runs; (4) more experiments means more researcher degrees of freedom and post-hoc analysis, even if unintentional, or done by an agent; and (5) verification is typically done post-hoc, and does not cover all steps of the research lifecycle.

We present Reproducible Science[1], a declarative, continuous, machine-checkable provenance layer for research claims, throughout the whole research lifecycle. `prereg` freezes a plan before the run, `results` seals its inputs, records its outputs and binds numbers in the manuscript to the run that produced them, `citations` pins exact quotations inside cited sources, and `repro` checks every declaration against the artifact it names. Our Claude Code plugin records provenance at the time of experimentation and reports drift on every edit through hooks. Applying this contract to this paper[2] surfaced two errors, one of which had already gone stale twice. As agents take over more of the research lifecycle, the artifacts they leave behind are worth only as much as we can verify the claims inside them.

1 Introduction

Surveys of research agents name verification as the step that has not kept pace [Ding et al., 2026]. Re-running the analysis is the usual answer, and it is usually unavailable: across 863,878 attempted executions of valid notebooks, 24.11% ran without errors and 4.03% produced the results the notebook recorded [Pimentel et al., 2019]. The deeper problem is that re-running checks the wrong object. A notebook that re-executes perfectly can still sit beside a manuscript printing a value no output contains, because execution establishes that the code runs and says nothing about what the paper claims the code produced. Reproducibility and verification are different properties. A project can preserve its code, its data and its logs and still print a value no output contains, quote a passage that was never read out of its source, or call an analysis confirmatory after its plan changed.

We call what accumulates **verification debt**. Like debt in software, it is cheapest to detect when it is introduced and expensive to reconstruct afterwards. At the moment a value changes, the run that produced the old one is still on disk, the source still resolves, and the edit carries an author and a commit. Later the run has been overwritten, the source was fetched once without its address recorded, and which document a citation referred to is no longer recoverable from the citation.

[1] Package: https://github.com/elliottower/reproducible-science
[2] Paper code and artifacts: https://github.com/elliottower/reproducible-science-paper

Accepted at the AI for Science workshop (NeurIPS 2026).

Of 34 documented cases of retraction, correction, or misconduct, Reproducible Science's checks would detect the stated problem in 6 and report the evidence as unchecked in 8 more, including both Surgisphere COVID-19 studies (Appendix A).

Table 1: Three boundaries of a research project, the failure each admits when nothing checks it, and the comparison Reproducible Science performs. All three are decided the same way, by a digest of a file and a comparison at a location the author declares in it, so one checking engine serves them.

Boundary	**What goes wrong when it is unchecked**	**What Reproducible Science compares**
plan to run	a hypothesis or an analysis is rewritten after the outcome is visible	a run record against a plan pinned by digest
run to manuscript	a printed value drifts from the output it was read from	a declared address in a sealed result file
source to manuscript	a quotation or an identifier does not occur in the artifact cited for it	a passage against a content-addressed source

Most of what a reader would want checked is not mechanically checkable: whether a design answers the question, whether a conclusion follows. A narrow part is. A number printed in a manuscript either matches the results file it came from or does not, and a quoted sentence either occurs in the cited source or does not. That part can be settled without a human and without a model, at a cost small enough to run on every commit rather than once before submission.

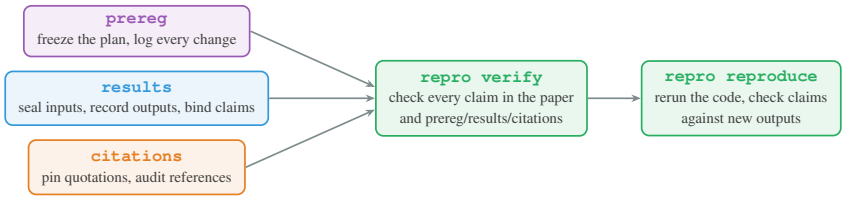


Figure 1: The commands of Reproducible Science. `prereg`, `results` and `citations` write declarations as the work is done; `repro verify` checks each against its artifact, and `repro reproduce` against rerun outputs.

Nothing failed because nothing ran. In the toolkit's own code, a missing extractor, the program that pulls text out of a PDF, returned the empty string, the check recorded an unreadable document, and the run summary read *nothing failed*. A check that read no text and a check that read the text and found it correct produced the same line. A separate incident is worse: a script collected the digests cited by every registration under one directory and verified each against a file; the directory was renamed, the glob matched nothing, the script exited zero, and continuous integration, a pre-push hook and a make target were all green in that state.

Position. Three positions follow, each against a live alternative. *Prospective decisions require prospective records*: a frozen plan cannot establish that an author was honest, but it makes subsequent rewriting detectable and separates confirmatory work from exploratory work. *Evidence requires declared addresses*: a reported `0.42` should name the pointer or selector it came from, because searching afterwards for a numerically similar value is not provenance and does not scale. The alternative is to recover the address afterwards, and a contemporaneous system does: Paperclip binds the same unit, a claim to a location in a source, and verifies it with a language-model subagent against line ranges into its own re-parse [Generative Expert Labs, 2026]. A verdict from a model is not recomputable from the artifact, and an address into a re-parse moves when the parser improves. We make the opposite choice on both: the comparison is deterministic, and an address points into pinned bytes. *Verification requires an explicit non-execution state*: a checker that cannot distinguish matched, disagreed, absent and could-not-run has no way to report that a green build examined nothing. Our own infrastructure violated the non-execution requirement twice.

Contributions. (1) **Lifecycle verification**: a model spanning the plan-to-run, run-to-manuscript and source-to-manuscript boundaries of Table 1, designed to run while a project changes. (2) **Ev-**

idence assertions: typed assertions binding manuscript values and quotations to declared locations in content-addressed artifacts, with four implemented variants over a shared engine and report. (3) **Staged outcomes**: a deterministic model separating execution, extraction and comparison, with a project's pass-or-fail policy applied to the resulting facts afterwards. We instantiate all three in a deployed toolkit (Figure 1) and evaluate them through conformance tests, fault injection, a quotation audit, recall against CORE-Bench, defects met in initial use, and an audit of a project we developed using it.

2 Model and Contract

Section 1's two incidents fail differently — one check never ran, the other ran and read no text — and both reported what a passing check reports. The three boundaries of Table 1 share one architecture: declare an address, pin the artifact it points into, compare deterministically, and report each stage separately. Two of the three are decided by an *evidence assertion*, which binds a manuscript claim to an artifact, a locator within it, and the value or passage expected there. A project lists its assertions in one file, the manifest. The plan-to-run boundary is decided by reading a run record beside a pinned plan, so no evidence assertion is defined for it. Four variants are implemented across the other two boundaries. They differ in how evidence is located and compared, and share artifact pinning, staged outcomes and one report. Appendix B provides the full definition of each term used here.

Table 2: The four implemented variants. A table cell is addressed by a selector over key columns, because a row index addresses a different cell when the table is reordered. A correspondence names no expected value: neither side is the reference, and a disagreement prints both readings.

Kind	Assertion	Located by	Compared how
metric	this artifact holds this value here	RFC 6901 JSON Pointer	decimal, at the precision the manuscript printed
table	this table holds this value in this cell	column plus a row selector	decimal, at the precision the manuscript printed
quote	this passage occurs in this artifact	page, optionally	verbatim after normalization that preserves which words appear
correspondence	two artifacts hold the same value	a locator on each side, prose anchors or a pointer	decimal, at the precision the manuscript printed

Three stages. A check must execute, extract, and compare, and the three fail independently. Table 3 gives the outcomes and the flattened view derived from them. The rows differ in what a reader should do: revise the claim, fix the pointer or run the analysis that produces the value, install a package, file a bug. A report that cannot distinguish them directs none of those actions.

Table 3: The three stages and the flattened view derived from them, the single label a report prints. A pointer to a key the results file lacks is reported `not_found`: nothing is asserted about whether the value differs. An extractor that is not installed yields `unchecked`, and an extractor that raises yields `error` and never `unchecked`.

Situation	Execution	Extraction	Comparison	Flattened
value present	completed	extracted	match	`verified`
value differs in a readable artifact	completed	extracted	mismatch	`mismatch`
pointer does not resolve	completed	absent	—	`not_found`
extractor not installed	unavailable	not attempted	—	`unchecked`
backend defect	failed	not attempted	—	`error`

Pins. Artifacts are content-addressed, and pins are resolved before any evidence is read, so a source that moved is known before decisions are computed against it. Decisions against a broken or unpinned artifact still run and are reported, marked non-authoritative, so a reader still sees what each assertion found.

Facts, then policy. Verification returns facts. Whether they constitute a pass depends on what the project is for: an unchecked citation is acceptable in a draft and disqualifying in a submission, and both readings are correct about the same run. The engine therefore reports the three stages and computes no verdict, and a profile maps outcomes to severities (Table 13, Appendix C).

3 Implementation and Workflow

The toolkit runs on ordinary files. It does not require an experiment to be conducted in a particular language, notebook system, analysis framework or agent runtime, and it does not require a manuscript to be authored inside its toolchain. The fifteen papers in the quotation audit of Section 4.2 were written without this contract in mind, and none was modified to be auditable. A value is read from nine formats, such as JSON and Parquet, and a quotation from five, such as PDF.

A new or existing project adopts the toolkit with one command, `repro init`. For full installation and usage instructions, see the online documentation.[3] A project declares as it goes. A plan is frozen before the run it governs, against a commit and a content hash. The script is committed, its inputs are sealed before an analysis and its outputs recorded after it, so every result can be regenerated: `repro reproduce` runs the declared command again and checks the manuscript's assertions against the file it writes. An edit to a reported value must then be matched by the committed script and the sealed inputs, and a reader who reruns the script sees any difference. For each sentence that prints a number, the manifest records the address the number came from, written when the sentence is. Verification reads the declarations and reports on each.

Table 4: The commands and what each establishes. The first three write declarations and `repro` reads them, applying a policy to the result.

Command	What it establishes
`prereg`	that the plan is unchanged since it was frozen, against a commit and a content hash
`results`	that a reported number came from a recorded run over sealed inputs
`citations`	that a quoted passage occurs in the source cited for it
`repro verify`	that every claim matches the file it came from, and that `prereg`, `results` and `citations` verify with no errors
`repro reproduce`	that every claim matches the new output when its command is rerun

A worked example. Appendix C states that the conformance suite holds 25 fixtures. The manifest `paper/worked_example/repro.yaml` in this paper's repository binds that sentence with two assertions: a `quote` assertion checking that the manuscript still prints it, and the `metric` assertion below, checking that the pinned inventory of the suite holds the number at the declared pointer.

```
- kind: metric
  artifact: conformance     # pinned: sha256 b9840b9d...
  pointer: /fixtures
  reported: '25'

$ repro verify --policy strict
  ok    conformance-fixtures quote      [short]
  ok    conformance-fixtures metric   /fixtures = 25
  2 verified
  policy strict: passed  (0 errors, 0 warnings)
```

With `reported: '19'`, the stale count of Appendix C, the line is marked `MISS`, the run exits 1, and the report reads *manuscript prints 19, conformance_inventory.json holds 25 at /fixtures*. The Claude Code plugin prompts for declarations while the artifacts are produced: after each edit it names any number or quotation in a manuscript that nothing yet backs, and before an analysis runs it says when the governing plan is unfrozen. A companion Claude Code mod shows the state of a project's records in the session, one row per command, with a warning when a frozen plan was edited or a run was recorded with nothing sealed (Appendix D, Figure 2).

Preregistration is optional and per claim. A claim is marked confirmatory, exploratory, or not applicable, and only the first requires a plan behind it. This matters for the honesty of the report: a

[3] https://reproducible.science

project that registers nothing is not failed for it, and a project that calls a claim confirmatory is held to the plan it named.

Cost is what makes the check a commit-time one. Verifying this paper's 1,062 assertions over 79 artifacts takes 1.28 seconds on a laptop and 0.55 seconds on a cloud container. Resolving its 203 quotations against 55 content-addressed sources takes 11.19 and 7.17 seconds when every source is read, and 1.90 and 0.94 seconds when the text extracted from an unchanged source is kept (Appendix E).

4 Evaluation

The conformance suite and the fault injections are in Appendix C: 25 fixtures, each recording its expected outcome, reason and artifact validity as a triple so a case cannot pass by reaching the right verdict for the wrong cause, and four injected faults each distinguishing a class the flattened view would otherwise merge.

4.1 Self-audit

This paper is an instance of what it describes. Each number its manifest binds carries two assertions: that the manuscript states the sentence, and that a pinned artifact holds the value at a declared pointer. The body binds 96 claims this way, and every claim in the manifest passes under the `strict` policy profile. Of the 1,020 numbers the PDF prints, appendices included, 533 are checked against the file they came from and the other 487 are set aside by a rule such as a year or with a stated reason. The manifest and the artifacts it pins are in the repository, so a reader can rerun the check from a clone. No claim here is confirmatory, so none requires a registered plan.

4.2 Quotation audit

Six extractors read the same 2,364 passages, quoted from fifteen published papers, out of the same pinned bytes and returned six different answers, from 2,314 down to 1,716 (Table 12). A verification snapshot therefore records the reader as much as the corpus, and a rerun under a different toolchain is not guaranteed to find the same issues. Under poppler's layout mode 113 quotations are not found and every failure traces to extraction (Appendix C), so a checker with only pass and fail would have accused the manuscripts 113 times with no misquotation present.

4.3 Recall against CORE-Bench

The corpora above carry no ground truth: nobody has labeled which of a paper's numbers its artifact really holds, so a located value is evidence of an occurrence. CORE-Bench provides that ground truth: each capsule, CORE-Bench's package of a paper's code and data, names the values its analysis produces, so recall is measured against a stated answer set rather than estimated [Siegel et al., 2024].

Over 45 capsules the benchmark states 240 values, of which 146 are located in the capsule's files without running it: a recall of 61%. Table 5 gives it by the strength of the value, where strength is the number of constraining digits. Recall falls as values carry more digits and then rises again, because short integers occur by chance in most artifacts and long decimals rarely do. A second run on a separate machine produces a byte-identical record for all 45 capsules.

5 Findings from Deployment

5.1 Defect classes surfaced in initial use

Pinning and checking our own projects with the toolkit surfaced five defect classes. Table 6 lists them. Each is reported as a class that occurs and is mechanically detectable; no frequency is claimed, and none of these counts enters any rate in Section 4. One of the five is a defect in a check rather than in a manuscript, which is the case the three-stage model exists for.

Two that integrity checking cannot reach. In the first row of Table 6 every artifact resolves at its recorded digest, so no integrity check fires: the files hold the values the run wrote, and the

Table 5: Recall over 45 CORE-Bench capsules, against the values each capsule states it produces. Eight capsules hold a record no reader here opened, so their misses are `unchecked` rather than `absent`: a two-state report would count all of them against the capsule. This is the same separation Section 4.2's quotation audit makes, on a third party's corpus and against a known answer set.

Strength	Stated	Located	Recall
trivial	59	54	92%
weak	76	40	53%
moderate	44	15	34%
strong	61	37	61%
all	240	146	61%

Table 6: Five defect classes surfaced by running Reproducible Science over one author's corpus. Status records whether a change has been made that prevents the class, not whether the instance was repaired.

Class	What is observably wrong	Status
stored value disagrees with the printed value	nine of ten table rows disagree with the file the row is read from	open
recorded hash compared against nothing	digests recorded and never compared to the bytes on disk	closed
check that runs on no files	a glob matches nothing, the check exits zero, and the build is green	open
assertions separated from the evidence they name	178 assertions marked verified, and no source in any commit	closed locally
reported figure that stopped reproducing	a reported count no longer reproduces from the bibliography it described	open

manuscript's table holds different values. A value printed at `0.001` is stored as exactly zero, and one printed at `0.0003` as `9.73e-7`. The disagreement is reachable only by a check that compares a file to the sentence citing it. In the fourth, a repository split carried 178 assertions and left the sources they named behind; the stored `verified` marks survived the split, so the repository reported a fully verified set of assertions that a reader could recheck in no part. Quotation verification had passed over every one of those sources, because a quotation resolves against the text as it stands.

5.2 What a reader can recheck from a clone

One command, `repro audit`, runs the toolkit over a clone of the public repository of a project we developed using it, at a pinned commit.[4] The project keeps registered plans, a results ledger and pinned quotations, and declares no evidence manifest. The clone carries 1,267 tracked files, three registered plans, a ledger of 153 events binding 131 manuscript claims to 13 runs, and seven claim records holding 18 quotations. Table 7 gives what each command established and could not.

The quotations do not verify from a clone: a publisher's PDF is not the author's to redistribute, so the repository carries only each source's digest, and all 18 quotations come back `unchecked`.

In a preregistered study of drug-trial outcomes, four rounds of journal review raised 45 comments, mostly on design, presentation and method, and our checks reported 31 further findings, mostly on values and references (Appendix I).

A certificate of reproduction does not make a paper's numbers checkable. Of 133 certified reproductions in the CODECHECK register [Nüst and Eglen, 2021], 6 leave a file holding one of the paper's tables, and under half the numbers in those tables have a cell to check against (338 of 713). Of the 338, 326 match the certified file up to rounding or file version, and 12 differ (Appendix F).

In a literature census we conducted with a language model and coding agents, Reproducible Science checked every extracted sentence. The model screened 43,462 abstracts and extracted 14,053 records, each carrying the sentence it was drawn from. Read by `citations verify` against the stored abstract, 13,989 match exactly, 27 match in two places, and 37 leave out one word (Appendix N).

[4] https://github.com/elliottower/knockout-epistasis-dynamics, commit `cb82384`.

Table 7: The four commands of Reproducible Science over a clone of the project's public repository. Two establish their result from the clone alone, because the plans and the ledger are committed with it. The quotation check reads no source and reports that, where a two-state report would have to print a pass or a failure.

Command	What it found	What it could not establish
`prereg`	three plans, each unchanged against the digest recorded when it was frozen	whether a plan was written before the outcomes it governs were seen
`results`	the chain is intact and anchored at 153 events; 8 of 12 recorded files match their digest, and 4 carry more than one recorded digest	what share of the manuscript's numbers the 131 claims cover: the manuscript is not in the repository
`citations`	quotations: 18 against 7 sources, none checked; bibliography: 28 of 32 entries checked, 6 differ from their registry record	whether any quotation occurs in its source
`repro`	no evidence manifest; exit 2, nothing addressed	any assertion, none having been declared

5.3 An edit to a registered plan

A preregistration carries a recorded digest, and an edit to it is detected by recomputing that digest rather than by asking the author what changed. In our own use, `prereg check` caught a subagent that had edited a frozen plan partway through an analysis, and the edit was reverted. The Claude Code plugin and mod refuse an agent's file edits to a frozen plan, and any other edit fails `prereg check`. Changes go in a separate amendment, sealed the same way.

6 Related Work

Claim-to-location binding. Paperclip binds individual claims to individual locations in sources: an agent adds a claim naming a document identifier and a line range, and a commit verifies each claim against the source text, dropping unsupported claims, with a log for audit [Generative Expert Labs, 2026]. The unit is the same as ours and three properties differ. Verification is performed by a language-model subagent, so the verdict is not reproducible from the artifact. Line ranges address the system's own parse of a document, which is refetched and re-read, so an address points into a mutable representation and improving the parse moves every stored range. No digest is recorded, which under Section 2 leaves every such artifact unpinned and every decision against it non-authoritative. A commit also lands when verification is skipped, which is the condition Section 1 describes. SciTeX Clew traces a registered claim back through the hashes of the scripts and inputs behind its source file [Watanabe, 2026]; the claim's value is stored but not compared with that file, which is hashed whole, without a location inside it.

Repository auditing. adduce audits a research repository against 78 rules and returns a score, asking whether "for every number in the paper, can I point to the artifact that produced it, and will that artifact still produce it elsewhere?" [Chudasama, 2026], which is this paper's question reached independently. A rule there returns one of five statuses, and two of them, `unknown` and `not-applicable`, are excluded from the score rather than counted as passes. The unit differs, and so does where an outcome comes from. adduce grades a repository and a rule author selects among five flat statuses, where one assertion here binds one printed value to one declared address, no score is computed, and the outcome follows from which of the three stages stopped. The two compose: adduce runs `repro verify` as one of its rules.

Artifact-level verification. Code Ocean verifies a capsule before publication and states that the check establishes mechanical reproducibility, a user pressing run and obtaining results, and separately that it does not check whether a capsule's results match the article it accompanies [Code Ocean, 2026]. Whole Tale packages code, data, methods and environment with the narrative as a re-instantiable tale [Brinckman et al., 2019]. In both, the unit of citation and the unit of verification are the same object, so the outcome is one property of the whole. Here the unit of verification is finer than the unit of citation, and a report states which of a manuscript's values an artifact still supports rather than whether the artifact runs.

Re-execution. Workflow systems rerun the steps whose inputs changed [Iterative, Inc., 2020, Köster and Rahmann, 2012], and a codechecker regenerates some or all of a paper's results [Nüst and Eglen, 2021]. ScientistOne ties each numeric sentence of a paper it writes to a line of its experiment log [Meng et al., 2026]. `repro reproduce` compares each printed number with the value at its declared address in the rerun output, and reports a rerun that did not happen apart from a rerun that disagreed (Appendix K).

Entailment, markup and write-time binding. FEVER, SciFact and CliniFact predict a label given retrieved text and score it against human annotation [Thorne et al., 2018, Wadden et al., 2020, Zhang et al., 2025], and ClaimReview binds claim text to the work it appeared in while recording no digest and enumerating no outcomes [Schema.org, 2026]. Our outcomes name what a byte or decimal comparison found. Literate programming generates the document from the execution producing its values [Leisch, 2002], which removes the drift, requires the manuscript to be authored inside the toolchain, and regenerates a changed value without reporting that it changed.

Abstention as an outcome. Fellegi and Sunter classify a candidate pair as match, non-match, or possible match held for review, and prove that rule minimizes the review region at fixed error bounds [Fellegi and Sunter, 1969]. The three-way structure, with abstention as a first-class outcome rather than a failure, is the shape Table 3 takes. Their third state is reached probabilistically from estimated agreement patterns; ours is reached because an extractor was absent or a pointer did not resolve.

Verification in other fields. Metrology defines verification as evidence that specified requirements are met, and keeps verification apart from validation, which asks whether those requirements suit an intended use [Joint Committee for Guides in Metrology, 2012]; Reproducible Science performs verification, which relocates a reader's dependence on its authors [Hardwig, 1985].

7 Limitations

The extractor is trusted. Content addressing establishes that an artifact's bytes did not change and nothing about whether the extractor read them correctly. A layout parser that splices a value from the wrong column, or an optical pass that reads `0.9489` as `0.9439`, leaves the digest intact and returns a confident answer. The blockchain literature calls this the oracle problem [Albizri and Appelbaum, 2021]. Recording the extractor's identity and a digest of the text it produced makes a changed reading visible, which addresses drift and not correctness.

Registration establishes order, not honesty. A run record states when a plan was registered and when the run began, and the check establishes that those two are consistent. It does not establish that the registration is contemporaneous with what it claims. Each freeze requests a date from outside the repository [Adams et al., 2001]: its digest goes to OpenTimestamps [Todd, 2016] and the returned proof fixes the latest date the plan could have been written (Appendix G). A timestamp cannot show that no analysis was run before the plan was frozen.

Declaration is authored, and coverage is gameable. Only declared numbers are checked, so a project can pass by declaring only the numbers it expects to match. We state coverage alongside outcomes so that a reader can see what fraction of the manuscript's numbers was checked. The records are written by the author. They make unintended drift visible to a reader with a clone; they do not resist an author or agent who rewrites the records themselves.

8 Conclusion

Re-running an analysis establishes that the code runs. Verification debt accumulates in the difference between that and what a manuscript says it produced. Values, quotations and registered plans differ in how each is extracted and little else, so one engine reports all three. The three stages exist because disagreement, absence and non-execution have different remedies, and because collapsing them produced a run that examined nothing.

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A Retractions, errors and misconduct in published results

The toolkit checks that a manuscript agrees with its artifacts, which covers some of the reasons published results are retracted and not others. To determine which reasons the checks cover, we examine 34 cases of retraction, correction, or misconduct finding, and classify each by the cause stated in its primary record.

Detected failure modes. Table 8 gives the coverage of each failure mode, counting each case once under its principal cause. The checks cover a stated problem in 14 of the 34 cases: the principal cause in 8 (6 detected and 2 reported as unchecked), and missing data reported alongside the principal cause in 6 others. The checks detect three failure modes directly: (1) a printed value that disagrees with the authors' own records fails the check of that number against its results file, (2) a report that departs from its registered plan differs from the frozen and timestamped plan, and (3) a reference that does not exist cannot be pinned to a source.

Table 8: Failure modes among the 34 cases and the toolkit's coverage of each. Each case is counted once, under its principal cause. Three of the ten modes defined before any record was read have no case here and are not shown: a result altered after the run, selective reporting with no registration to compare against, and a method that is invalid though correctly implemented. †Added after the records were read.

Failure mode	Cases	Coverage
printed value disagrees with the authors' own records	3	**detected**: the printed number fails the check against its results file
report departs from the registered plan	1	**detected**: the report differs from the frozen, timestamped plan
references that do not exist	2	**detected**: a reference that does not exist cannot be pinned to a source
data could not be produced for audit	2	**reported as unchecked**: the sealed digest records that the file existed
error inside the analysis	9	**not detected**: the sealed script and inputs allow the analysis to be rerun
data fabricated or falsified	8	**not detected** if fabricated before sealing; a later change to a sealed file is detected
image manipulation or duplication	4	**not addressed**
study conducted differently from its description†	4	**not addressed**
none of the above	1	**not addressed**
Total	34	

Undetected failure modes. Data that cannot be produced for an audit are reported as unchecked rather than verified. An error inside the analysis passes every check, because the manuscript agrees with the output of the faulty analysis, but the sealed script and inputs allow a reader to rerun the analysis and locate the error. The toolkit does not address data fabricated before sealing, manipulated images, or a study conducted differently from its published description.

Case selection. The primary record of a case is its retraction notice, a finding of the United States Office of Research Integrity, or a published reanalysis. The cases comprise nine of the ten most-cited retracted papers listed by Retraction Watch [Retraction Watch, 2026], publications named in Office of Research Integrity findings, retractions drawn from the Retraction Watch dataset [Crossref and Retraction Watch, 2026] by reason tag, and well-known examples such as the Duke chemotherapy predictors and the Surgisphere studies. We select the cases as examples of each failure mode. We define ten failure modes before reading any record and add an eleventh for four records that fit none of the ten.

Retraction Watch dataset. To place the cases in context, we analyze the reasons recorded in the Retraction Watch dataset, which covers 67,530 retractions. We exclude the 24,166 retractions attributed to paper mills or compromised peer review, because these concern the publication process,

and group the reason tags of the remaining 43,364 (Table 9). Among these, 4,273 retractions (9.9%) cite an error in the analysis, data, methods, results or text, and 1,909 (4.4%) cite original data or images that were not provided. Retractions that cite unavailable data or images follow publication by a median of 4.91 years, compared with 1.31 years for retractions that cite an error.

Table 9: Retractions in the Retraction Watch dataset by group of reason tags, among the 43,364 not tagged for paper mills or compromised peer review. A retraction can fall in several groups. Years run from the paper's date to the retraction's date.

Reason for retraction	Retractions	Share	Median years to retraction
image duplication, manipulation, fabrication or concerns	7,171	16.5%	4.39
error stated in analysis, data, methods, results or text	4,273	9.9%	1.31
original data or images not provided	1,909	4.4%	4.91
falsification or fabrication of data	1,659	3.8%	4.27
results not reproducible	861	2.0%	2.01
limited or no information in the notice	11,957	27.6%	0.11

Cases. Table 10 gives the cause quoted from the primary record of each case.

Table 10: Thirty-four cases, grouped by failure mode, with the cause in the words of the primary record: a retraction notice, a finding of the Office of Research Integrity, or a published reanalysis.

Case	What its primary record says
Printed value disagrees with the authors' own records	
Abegaz et al. (2014), PLOS ONE	"the estimated number of deaths and injuries reported cannot be reproduced when using the unmatched and matched police and hospital data" [The PLOS ONE Editors, 2026a]
Sun et al. (2024), PLOS ONE	"the p50 median value is lower than the reported minimum value" [The PLOS ONE Editors, 2026g]. Also states that data, images or records could not be produced.
Wansink et al. (2012), Archives of Pediatrics and Adolescent Medicine	"We had erroneously reported the age group as children ranging from 8 to 11 years old; however, the children were 3 to 5 years old." [Wansink et al., 2017]
Report departs from the registered plan	
67 trials in five medical journals, 2015 to 2016 (a cohort, not one publication)	"365 novel outcomes were reported without declaration in total, a mean of 5.4 per trial" [Goldacre et al., 2019]
References that do not exist	
Khalilzadeh et al. (2026), PLOS ONE	"Citation of unretrievable references 1, 2, 4, 5, 8, 10, 12–18, 20–22, 33, 36, 51, 61, 75, and 79." [The PLOS ONE Editors, 2026c]
Mohammad et al. (2026), PLOS ONE	"Citation of unretrievable References 2, 3, 6, 7, 8, 18, 20, and 26." [The PLOS ONE Editors, 2026e]
Data could not be produced for audit	
Mehra et al. (2020), Lancet	"Surgisphere would not transfer the full dataset, client contracts, and the full ISO audit report to their servers for analysis" [Mehra et al., 2020b]
Mehra et al. (2020), New England Journal of Medicine	"the raw data could not be made available to a third-party auditor, we are unable to validate the primary data sources" [Mehra et al., 2020a]
Error inside the analysis	
Morisky et al. (2008), Journal of Clinical Hypertension	an independent statistical review "concluded that the results were misleading due to issues regarding the sensitivity and specificity of the medical adherence scale used" [Wiley Periodicals LLC, 2023]
Gebrechorkos et al. (2025), Nature	"area-weighted grid cells were not used when calculating study-wide averages, an error that affected numerical values in the main text" [Gebrechorkos et al., 2026]
Islam et al. (2019), PLOS ONE	"an error in the p-value calculation caused by a coding error" [The PLOS ONE Editors, 2026b]

Case	What its primary record says
Zhang et al. (2025), PLOS ONE	a formula "contains an incorrect minus sign, which should be a plus sign" [The PLOS ONE Editors, 2026h]
Okray et al. (2023), Nature	the authors found "errors in the pipeline originally used for data analysis" [Okray et al., 2026]
Hewitson et al. (2021), PLOS ONE	the correlation analysis "included typically developing (TD) participants in error" [The PLOS ONE Editors, 2024]
Rahman et al. (2021), PLOS ONE	"the PSS mean reported in [1] is consistent with incorrect coding of these items" [The PLOS ONE Editors, 2026f]. Also states that data, images or records could not be produced.
Huang et al. (2021), Lancet	an expression of concern was published "because of the substantive data correction request from the authors following comments from a reader" [The Lancet Editors, 2023]
Reinhart and Rogoff (2010), American Economic Review	"coding errors, selective exclusion of available data, and unconventional weighting of summary statistics lead to serious errors" [Herndon et al., 2013]
Data fabricated or falsified	
Potti et al. (2006), Nature Medicine	"Respondent altered data sets to improve the accuracy of predictors for response to treatments" [Office of Research Integrity, 2015]; "the results incorporate several simple errors that may be putting patients at risk" [Baggerly and Coombes, 2009]; "a corruption of several validation data sets precludes conclusions regarding these signatures" [Potti et al., 2011]
Potts Kant et al. (2013), Pediatric Research	falsified data by "changing numerical values in the datasets generated by the apparatus" [Office of Research Integrity, 2019]
Savine and Braver (2012), Cognitive, Affective and Behavioral Neuroscience	"Respondent exaggerated (1) the effect of incentive context on response times and error rates in Table 1" [Office of Research Integrity, 2013]
Skau et al. (2016), Cell	falsification by "selectively reporting by inappropriate inclusion/omission or alteration of data points in ten (10) figures" [Office of Research Integrity, 2018]
Hauser et al. (2002), Cognition	"Respondent published fabricated data in Figure 2 of the paper" [Office of Research Integrity, 2012a]
Linn et al. (2003), Cellular and Molecular Biology	"he falsified the values in Tables 2-5 by altering the values that he had obtained from another author's manuscript" [Office of Research Integrity, 2010]
Thiruchelvam et al. (2005), Journal of Biological Chemistry; Rodriguez et al. (2005), Environmental Health Perspectives	misconduct by "falsifying and fabricating cell count data that she claimed to have obtained through stereological methods" [Office of Research Integrity, 2012b]
Taylor and Gercel-Taylor (2008), Gynecologic Oncology	two investigations "found evidence of falsification, fabrication, and plagiarism of data" [Taylor and Gercel-Taylor, 2023]
Image manipulation or duplication	
Jiang et al. (2002), Nature	"The original images for Figures 6a, 6e and 6m could not be retrieved by the authors" [Jiang et al., 2024]
Lesné et al. (2006), Nature	the figures "show signs of excessive manipulation, including splicing, duplication and the use of an eraser tool. The data cannot be verified from the records." [Lesné et al., 2024]
Zou et al. (2001), Nature	falsified figures by "manipulating the images to alter the number and location of positively stained cells" [Office of Research Integrity, 2014]
Liu et al. (2014), PLOS ONE	"the data underlying the panels of concern are no longer available" [The PLOS ONE Editors, 2026d]
Study conducted differently from its description	
Wakefield et al. (1998), Lancet	the paper's claims "that children were 'consecutively referred' and that investigations were 'approved' by the local ethics committee have been proven to be false" [The Editors of The Lancet, 2010]
Estruch et al. (2013), New England Journal of Medicine	"Because of irregularities in the randomization procedures, we wish to retract the following article" [Estruch et al., 2018]

Case	What its primary record says
Gautret et al. (2020), International Journal of Antimicrobial Agents	"The journal has not been able to establish whether all patients could have entered into the study in time" [Gautret et al., 2025]. Also states that data, images or records could not be produced.
LaCour and Green (2014), Science	"Survey incentives were misrepresented."; "LaCour has not produced the original survey data from which someone else could independently confirm the validity of the reported findings" [McNutt, 2015]
None of the above	
Fukuhara et al. (2005), Science	"we have found that not all preparations of visfatin bind to and activate the insulin receptor" [Fukuhara et al., 2007]

Limitations. The 34 cases are illustrations, at least one per failure mode in Table 8. Table 9 reports how often each reason for retraction occurs among 43,364 retractions in the Retraction Watch dataset. The reason tags reflect the dataset curators' reading of each notice, and 27.6% of the retractions analyzed are tagged as having a notice with limited or no information. In the full dataset, 1,725 of the 2,464 retractions tagged for unavailable data or images also have an image-related tag, so most concern missing original images. The analysis shows which failure modes the toolkit's records would expose; it does not show that any of these cases would have been corrected sooner.

B Terms

Table 11 defines the terms used in this paper.

Table 11: Terms as this paper uses them.

Term	Meaning
Reproducible Science (the toolkit)	the `reproducible-science` package, with its commands `prereg`, `results`, `citations` and `repro`; an MCP server, `repro mcp`, for other agents; and integrations with Claude Code, a plugin and a mod, and an experimental one with Codex (Appendix M)
engine	the code in `repro` that resolves pins, evaluates each assertion and reports the three stages
artifact	a file a claim rests on, pinned by its SHA-256 digest
claim	a sentence in the manuscript that states a number, a quotation or a cited finding
evidence assertion	one check bound to a claim: an artifact, a location in it, and the value or passage expected there
manifest	the file, `repro.yaml`, listing a project's artifacts, claims and assertions
claim record	a file pinning the quotations taken from one source, read by `citations`
plan	the preregistration document
freeze	the record of a plan's digest, commit and time, after which the plan does not change
ledger	the append-only record of sealed inputs, runs and the claims bound to them, kept by `results`
declaration	a frozen plan, a sealed input, a recorded run, a pinned quotation or an assertion: what a project writes for the toolkit to read

C Conformance suite and fault injection

Conformance. The suite holds 25 fixtures: five addressing a table cell, four comparing a value, three each for prose anchors, two-sided correspondences and artifact availability, two each for a pointer and a passage, and one each for an unpinned artifact, a broken pin, and a claim offering no evidence. Each records its canonical expected output as an (outcome, reason, artifact validity) triple, so a case cannot pass by reaching the right verdict for the wrong cause. Assertions over them include that an absent pointer yields `extraction=absent` and never `comparison=mismatch`, and that an injected `TypeError` yields `error` and never `unchecked`.

Self-audit. Building the manifest surfaced two defects in this paper's own numbers. Four counts in Table 12 had been computed correctly and written to no file, so nothing could re-check them; derived into an artifact, all four match what the manuscript prints. The count of conformance fixtures was recorded and stale twice: the specification stated eighteen against a count of nineteen, and a stored figure gave nineteen where the suite holds 25.

Fault injection. Four faults are injected and the reported outcome recorded: a results file edited after drafting yields a broken pin with decisions against it marked non-authoritative; an absent extraction toolchain yields `unchecked` with reason `extractor_missing`; a registered plan edited after registration yields a broken pin on the plan artifact; a backend raising `TypeError` yields `error` with reason `backend_defect`. Each distinguishes a class the flattened view would otherwise merge.

The quote variant was run over an audit of sixteen claims from published papers: 2,512 quotations against 125 content-addressed sources, of which the fifteen PDFs of the papers under audit carry 2,364. Every source carries a digest, so the document a failing quotation was checked against is byte-identical to the one the manuscript cited. Under poppler's layout mode, 113 quotations were not found. One rater, who also wrote the quotations, attributed every one to extraction rather than to misquotation, each cause being recoverable from the recorded extractor output: 110 came from the mode's preservation of page geometry, which severs every sentence crossing the gutter of a two-column paper, 2 from a text layer omitting a character the passage contains, and 1 from a hyphen at a line end that de-hyphenation cannot attribute. On this corpus a two-state checker therefore carries a false-accusation rate of 4.5%: it would have recorded all 113 as misquotations, each an accusation against the manuscript, and the corpus held no misquotation for it to find.

Checking the extractor. Authors should check the extractor against their own sources. The tools verify that a quotation occurs in the text an extractor produced, not that the extractor read the page correctly, and only someone who has seen the page can confirm that [Wagenknecht, 2014]. The citations tool lets authors record which passages they have checked against the page and, where an extractor misreads a passage, pin the source under another extractor or record a correction.

Table 12: Passages resolved per extractor over one corpus, and the cost of each. The two poppler modes fail in opposite directions and together reach 2,360. Ranking by total resolved is a poor guide to which extractors to consult: the low-level parser resolves 598 fewer passages than the best single extractor and is the only one that extends the union, reaching 2,362. Two quotations resolve under no extractor.

Extractor	Resolved of 2,364	Seconds
poppler, reading order	2,314	6.9
poppler, layout	2,250	6.9
pure-Python reader	2,229	43.9
rendering-engine binding	2,092	4.8
structural extractor	1,973	6.2
low-level parser	1,716	2.7
both poppler modes	2,360	13.8
all six	2,362	71.4

Table 13: Policy profiles over identical facts. Every profile requires that at least one assertion was evaluated, because a project with no evidence anywhere otherwise satisfies every condition trivially. `not_offered` marks a claim that offers no evidence.

Profile	mismatch	not_found	unchecked	not_offered	unpinned
exploratory	error	warning	warning	ignore	ignore
publication	error	error	warning	warning	warning
publication, confirmatory claims	error	error	**error**	**error**	—
strict	error	error	error	error	error

D Session status

The Claude Code mod reads the project's records at the end of each turn and draws them above the prompt, one row per command: the plans frozen out of the plans present, the runs sealed and the claims bound to them, the quotations pinned, and the manifest's checks verified. A row for a record the project lacks names the command that creates it. A line appears under the prompt only when something needs attention: a record that no longer matches, runs recorded with nothing sealed or no plan frozen, or a fault in the ledger.



Figure 2: The state of a project's records as a session shows it. One plan is frozen, one run is sealed with one claim bound to it, no quotation is pinned, and three of the manifest's four checks verify. The line under the prompt reports the fourth: a number in the manuscript no longer matches its results file.

What is automatic. The plugin's hooks write no file: they refuse an edit to a frozen plan, warn before an analysis runs under an unfrozen plan, and after an edit report an added number no ledger claim names, an added quotation no claims file pins, or a frozen plan that changed. The mod's end-of-turn check records changed outcomes inside the git directory. The researcher or agent writes every declaration: a printed number is backed by `results seal` on its inputs, `results run` on its output, `results claim` on its sentence, and an assertion in `repro.yaml` that `repro verify` checks. Given result files, a manuscript and no manifest, `repro init` drafts claims with no assertion in `repro.draft.yaml`. Each `repro mcp` proposal is inferred and unverified until a person accepts it.

E Timing

Each time in Section 3 is the median of 10 runs of the command a user types, each in a new process, with version 0.6.0 on two machines: a 2019 16-inch MacBook Pro (`MacBookPro16,1`: 6-core Intel Core i7-9750H at 2.6 GHz, 16 GB of memory, macOS 26.5, Python 3.13.12) and a Modal container (2 processors and 4,096 MB of memory requested, Linux under gVisor, Python 3.12.10). Both read the same manifest and the same pinned sources. PDF text is extracted by poppler's `pdftotext`, version 26.04.0 on the laptop and 22.12.0 in the container. The laptop's load average over the previous minute was 3.37 when its runs began, and its slowest run reading every source took 14.287 seconds. By default `citations verify` keeps the text extracted from a source whose bytes have not changed; Table 14 times it both ways.

Table 14: Seconds to check this paper's records, over 10 runs on each machine.

Command	Machine	Median	Fastest	Slowest
`repro verify`	laptop	1.28	1.230	2.578
	container	0.55	0.545	0.573
`citations verify`, every source read	laptop	11.19	10.912	14.287
	container	7.17	6.994	7.421
`citations verify`, extracted text kept	laptop	1.90	1.876	1.993
	container	0.94	0.924	3.098

Table 15 gives `repro verify` over generated projects of increasing size, beside `sha256sum -c` over the same artifacts, the least any check of pinned files can cost. Each project holds a text, a table and a manifest of correspondence assertions, each tying a number in the text to a cell of the table. The command starts in 0.192 seconds, which is most of its cost up to 100 assertions. Artifact size adds what hashing adds: over 4 GB the check takes 2.416 seconds and hashing alone takes 2.163. Time grows faster than the number of assertions past 10,000, because each text anchor is searched for across the whole text.

Table 15: Seconds to verify a generated project, by the number of assertions and by the bytes of the artifacts they are checked against. Each time is the median of 5 runs, each in a new process, of version 0.5.2 on a GitHub-hosted runner (`ubuntu-24.04`: AMD EPYC 9V45, 2 processors, 7 GB of memory, Python 3.12.15). The last four rows hold the assertions at 100 and add pinned files that no assertion reads.

Assertions	Artifact bytes	`repro verify`	`sha256sum -c`
10	647	0.213	0.001
100	6,389	0.238	0.001
1,000	63,807	0.556	0.001
10,000	638,009	6.733	0.002
100,000	6,380,011	193.128	0.005
100	1,054,965	0.223	0.002
100	16,783,605	0.246	0.010
100	268,441,845	0.372	0.142
100	4,294,973,685	2.416	2.163

F Certified reproductions

CODECHECK issues a certificate when an independent codechecker regenerates a paper's outputs, and each certificate's manifest names the regenerated files and the figure or table each belongs to [Nüst and Eglen, 2021]. The correspondence between file and table is therefore written by a third party. We run `repro verify` over the public register to measure how much of a certified reproduction can be checked number by number. The plan is frozen before any output file is opened.

Result. Six certificates link fifteen tables to regenerated files, and 338 of the 713 numbers those tables print are bound to a cell of a file. In 326 of the 338, the paper and the certificate's file hold the same number. Twelve differ, in two certificates, and each difference has a cause that can be read from the files (Table 19). The counts describe what can be checked in certified reproductions and do not estimate how often published numbers are wrong.

From the register to checkable tables. Table 16 follows the register down to the certificates the toolkit can use. The manifests name 545 outputs, of which 419 are figures. Six certificates name a file the toolkit reads, keep it in their repository, and say which table of the paper it holds.

Table 16: Certificates in the CODECHECK register that reach each condition. Three certificates describe their files without naming a table, and four name a file their repository does not hold under that name.

condition	certificates
in the register	133
names an output in a format read as values, not marked as outside the manuscript	13
that output is in the certificate's repository	9
its manifest comment names a table the paper has	6

Coverage. A number is bound when it has one address in the paper's extracted text and one cell in the regenerated file. The bindings are written with Claude Code (Anthropic) from views of each output in which every value is masked, and are locked before any comparison runs. Four certificates bind between 64 of 92 and 137 of 140 numbers (Table 17). Two bind none: their outputs hold formatted text, in one a median with its interval and units in each cell and in the other a single

column named `text`. An output that reproduces a table's appearance has no cell to compare a number against.

Table 17: Numbers printed in linked tables, counting at most 60 per table, numbers bound on both sides, and bound numbers that are the same in the paper and in the certificate's file.

certificate	printed	bound	same number	differ
2025-018	140	137	133	4
2020-025	90	86	86	0
2022-001	92	64	56	8
2024-020	58	51	51	0
2020-011	33	0	—	—
2020-008	300	0	—	—
all	713	338	326	12

Agreement. The toolkit reports 254 of the 338 bound numbers as agreeing at the precision the paper prints. It reports 69 as disagreeing and does not read 15, and 72 of those 84 are the same number (Table 18). The frozen rule reads each output at the last commit before the certificate's check time. Only sixteen of 37 linked outputs exist at that commit, so before any output was opened we amended the rule to read each output at the first commit holding it. The 30 numbers of one table of certificate 2020-025 differ from that first revision of its output file and equal the revision at the check time in every cell; the certificate does not record which revision was certified. The 15 unread numbers, 11 of them equal to the file, expose a defect in the toolkit, whose numeric parser rejects a thousands separator.

Table 18: The 338 bound numbers, by how the paper's numeral compares with the certificate's file.

comparison	numbers
the check reports agreement	254
the paper truncates where the check rounds	31
equal at the output file's revision at the check time	30
equal once a thousands separator is removed	11
differ	12
all	338

Differences. Table 19 lists the 12 numbers that differ. Five are standard deviations that equal the file's value multiplied by $\sqrt{(n-1)/n}$ with $n = 1{,}070$, which is consistent with a paper that divides by n where the regenerated file divides by $n-1$; the same factor reproduces all 12 standard deviations counted in that table. One regression coefficient is printed without its exponent. Three are memory measurements, 0.1% to 7.0% from the file's value.

Planted changes. Agreement means little if the check cannot report a disagreement, so each bound number is checked again on copies carrying one planted change (Table 20). The check reports a one-unit change in the last printed digit of the file in all 254 cases and of the paper in 240. In ten of the fourteen misses the file stores fewer digits than the paper prints, and in four the changed numeral sits among identical neighbors that serve as its anchors. No deleted row, column or file is reported as agreement or as a disagreement.

Table 19: The 12 bound numbers that differ between the paper and the certificate's file. File values are shortened to the digits needed to show the difference.

certificate	quantity	paper	file	difference
2022-001	standard deviation	30.33	30.3485	divisor n or $n-1$
2022-001	standard deviation	21.80	21.8124	divisor n or $n-1$
2022-001	standard deviation	28.20	28.2116	divisor n or $n-1$
2022-001	standard deviation	84,154.56	84,193.92	divisor n or $n-1$
2022-001	standard deviation	3,132.90	3,134.36	divisor n or $n-1$
2022-001	regression coefficient	5.273	5.273×10^{-7}	exponent not printed
2022-001	median	4,984.44	4,984.4349	last printed digit
2022-001	maximum	827,067.40	827,067.43	0.03
2025-018	run time, seconds	12.659	12.658475	last printed digit
2025-018	memory, megabytes	3.1	2.897	7.0%
2025-018	memory, megabytes	610.8	611.443	0.1%
2025-018	memory, megabytes	2310.8	2329.968	0.8%

Table 20: Planted changes on copies of the bound numbers, and how many the check reports with the outcome and reason its specification gives. The first three rows use the 254 numbers that agree; the last three use all 338.

planted change	expected report	as expected
none	agreement	254 of 254
last printed digit of the file's value	disagreement	254 of 254
last printed digit of the paper's value	disagreement	240 of 254
addressed row deleted	row absent	338 of 338
addressed column deleted	column absent	338 of 338
output file deleted	not checked, file missing	338 of 338

G Outside timestamps

`prereg freeze` takes one SHA-256 digest over the plan and sends that digest, and nothing else, to the OpenTimestamps calendar servers [Todd, 2016]. The proof they return is kept beside the plan and committed with it. A proof is pending until the calendars commit it into a Bitcoin block, whose date neither the author nor a calendar operator can change, usually within hours; `prereg timestamp` then completes it and reports the block and its date, the latest date at which the plan could have been written. `prereg check` reads the proof without the network and fails when the proof is of a digest other than the frozen one. No option skips the request: a freeze made with no network succeeds and reports the timestamp as owed on every check until it is made. `results timestamp` dates the head of the ledger the same way.

H Reproducible Science Audits

This appendix audits 3 preregistered studies conducted with the toolkit (Appendices I, J and N): one accepted by a journal, one under journal review and one in progress. Each audit gives a description of the study, an example finding, what each command found in its repository, and the findings made while it was written. Table 21 gives the result from a clone for the two submitted to journals, and Table 22 their findings.

Table 21: What the toolkit established from a clone of each project, at the pinned commit.

project	appendix	registered plans	quotations	ledger
Cross-design evidence discordance	I	4 of 6 registrations unchanged, 1 appended, 1 changed	175 of 175 found	not in the repository
Knockout epistasis	J	3 of 3 plans unchanged	18 declared, sources absent	chain intact, 153 events

Findings are coded from each project's records: commits, ledger events, plan logs, claim files and session transcripts. A finding counts when a command, a script that performed the same check before the command existed, a hook, or an act the toolkit requires (sealing an input, binding a number, pinning a quotation) exposed it and a record says so. The coding rules were fixed before the records were read. Defects found by a person or raised in review are not counted as findings.

Table 22: Findings, by project and severity. A *claim* finding would change a reported number or conclusion. A *source* finding is a value or reference attributed to a work that does not contain it. A *record* finding is a break in provenance that leaves every reported number correct.

project	appendix	claim	source	record	cosmetic	total
Cross-design evidence discordance	I	5	14	8	4	31
Knockout epistasis	J	3	2	10	3	18
total		8	16	18	7	49

I Audit: Cross-design evidence discordance

Description. A preregistered study of drug-trial outcomes [Tower, 2026a].[5] The clone carries 365 tracked files, 6 registration documents, 94 claim records, 175 quotations and 75 bibliography entries.

All Findings. 31: 5 claim, 14 source, 8 record, 4 cosmetic.

Example Finding. `results verify` found two recorded files edited since they were recorded and one that had moved.

Reviews. Of the 31 findings: 0 before submission, 11 review round 1, 2 review round 2, 0 review round 3, 17 review round 4, 1 after round 4.

Table 23: Cross-design evidence discordance: what each command found in a clone at the pinned commit, and what it could not establish. Exit 0 is a pass, 1 a finding, and 2 nothing to read.

command	exit	what it found	what it could not establish
`prereg`	1	6 registration documents, each frozen by a commit recorded in the document: 4 unchanged, 1 with text added after the frozen text, and 1 changed after its freeze	whether a registration was written before the outcomes it governs were seen
`results`	2	no ledger	which run any reported number came from
`citations`	0	175 of 175 quotations found in their sources; 67 of the 94 sources were read from the library and matched to the SHA-256 the record pins	whether a quotation supports the sentence that cites it
`citations audit`	0	75 bibliography entries, 74 carrying an identifier; 64 agree with the record the identifier resolves to and 10 disagree	whether the entry with no identifier is correct
`repro`	2	no evidence manifest	any assertion, none having been declared

Table 24: Reviewer comments (4 reviewers, 6 reports) and findings for the cross-design study, by kind.

Reviewer comments	Count
design and argument	20
reporting and presentation	12
statistical method	10
references	1
code and data availability	0
other	2
Total	45

Findings	Count
value or citation absent from the cited source	9
value differing from its source	6
wrong reference metadata	11
number bound to no run	2
other record findings	3
Total	31

[5] https://github.com/elliottower/cross-design-evidence-discordance, commit `c6a7a06`.

J Audit: Knockout epistasis

Description. A study of knockout epistasis in Boolean gene regulatory networks, also covered in Section 5.2 [Tower, 2026b].[6] The clone carries 1,267 tracked files, 3 registered plans, 153 ledger events, 13 runs, 131 manuscript claims bound to a run, 7 claim records and 18 quotations.

All Findings. 18: 3 claim, 2 source, 10 record, 3 cosmetic.

Example Finding. Binding the manuscript's numbers to runs required recomputing them, and the recomputation revealed that a property stated for all 28 networks holds only for 27.

Table 25: Knockout epistasis: what each command found in a clone at the pinned commit, and what it could not establish. Exit 0 is a pass, 1 a finding, and 2 nothing to read.

command	exit	what it found	what it could not establish
`prereg`	0	3 plans, each unchanged against the digest recorded when it was frozen	whether a plan was written before the outcomes it governs were seen
`results`	1	chain intact and anchored at 153 events; 8 of 12 recorded files match their digest and 4 carry more than one recorded digest	what share of the manuscript's numbers the 131 claims cover, the manuscript not being in the repository
`citations`	1	18 quotations declared against 7 sources, each carrying a SHA-256; none checked, the sources not being in the repository	whether any quotation occurs in its source
`repro`	2	no evidence manifest	any assertion, none having been declared

K Precedents for Re-execution

The `repro` manifest, `repro.yaml`, lists each command to rerun, the input files the command reads, and the numbers the manuscript prints from its output. `repro reproduce` reruns each listed command in a directory that holds only that command's declared inputs, and then compares every number the manuscript prints from the output with the value at its declared address. Table 26 sets each of its design choices beside the closest earlier system and states the difference. Most choices have a direct precedent. Three do not, in the sources read: an outcome for a rerun whose number could not be read, a label on a step that ran after an upstream failure, and a comparison of a printed value with a declared address in an output file that stays with the project.

[6] https://github.com/elliottower/knockout-epistasis-dynamics, commit `cb82384`.

Table 26: Design choices of `repro reproduce`, the closest precedent for each, and the difference. Rerunning is common and is specified in many ways; a mechanical comparison against the numbers a manuscript prints, with outcomes that keep a rerun that did not happen apart from a rerun that disagreed, is where the command departs from the systems before it.

Design choice	Closest precedent	Difference
Commands to rerun are declared in a manifest	DVC lists stages that wrap shell commands [Iterative, Inc., 2020]; Snakemake states rules from input files to output files [Köster and Rahmann, 2012]; ReproZip traces one execution [Chirigati et al., 2016]; many ClawBio skills write a replay script when they run [Corpas, 2026].	A regeneration is declared beside the claims that read its output.
Each command runs with only its declared inputs	ReproZip unpacks into a directory, a chroot, a virtual machine or a container [Chirigati et al., 2016]; Code Ocean, Whole Tale and Binder package an environment [Code Ocean, 2026, Brinckman et al., 2019, Jupyter et al., 2018].	A directory of declared inputs and no container: an input that was not declared is absent.
Success is that the printed numbers still hold	A codechecker judges, and identical results are not required [Nüst and Eglen, 2021]; ACM badges ask for agreement within a tolerance [Association for Computing Machinery, 2020]; CORE-Bench accepts answers inside an interval from three reruns [Siegel et al., 2024]; SEER-Repro accepts a relative error of at most 5%, or a difference of at most half a unit in the last decimal the paper reports [Liu et al., 2026]; DVC compares a dependency's contents with their previous version to decide whether a stage is rerun [Iterative, Inc., 2020]; ClawBio gives checksums of selected output files for a reader to check [Corpas, 2026].	Each number the manuscript prints is compared mechanically, and the bytes need not match.
Outcomes are kept apart	A study of notebooks counts separately those whose dependencies failed to install, those that failed in execution, those that finished with different results and those that finished with the same [Pimentel et al., 2019]; ReproScore leaves its outcome score undefined without execution evidence [Samuel et al., 2026]; SEER-Repro labels a perturbed analysis inconclusive, unavailable or not tested, and counts a reproduced value that is missing or unparseable as not reproduced [Liu et al., 2026]; CODECHECK has no process for a failed check [Nüst and Eglen, 2021].	Five outcomes are recorded for every regeneration, and one that was not rerun never reads as reproduced.
A rerun whose number cannot be read is `unchecked`	Paperclip fails a paper whose extracted value is invalid [Generative Expert Labs, 2026]; ScientistOne sorts its extraction misreads by hand [Meng et al., 2026].	The case has its own recorded outcome.
A step after a failure still runs, and is labeled	DVC can skip or run the dependents of a failed stage [Iterative, Inc., 2020]; Snakemake deletes the outputs of an incomplete rule [Köster and Rahmann, 2012].	The dependent step runs over the pinned copy, and its record says so.
Every invocation is appended to a log	Workflow Run RO-Crate packages a run [Leo et al., 2024]; AiiDA records each process in a provenance graph [Huber et al., 2020]; in-toto writes a signed link for each step [Torres-Arias et al., 2019]; DVC keeps the current state in a lock file [Iterative, Inc., 2020].	One append-only file beside the project holds what each invocation observed.
Checking never executes	SLSA verification compares a digest and expectations, and rebuilding is optional [Open Source Security Foundation, 2023]; an in-toto verifier checks signed records of steps [Torres-Arias et al., 2019]; DVC has a dry run [Iterative, Inc., 2020].	`repro verify` and `repro reproduce` are separate commands with separate guarantees.
A printed value is compared with a declared address in an output file	ScientistOne tags each numeric sentence with a log line and compares within 5% [Meng et al., 2026]; when the document is built, Sweave inserts the output of its code chunks and showyourwork regenerates each figure from its script [Leisch, 2002, Luger et al., 2021]; a CODECHECK manifest ties an output file to a figure by a comment [Nüst and Eglen, 2021].	The address is declared, resolved and compared, an ambiguous address is an outcome, and the record stays with the project.

Chain-of-Evidence. ScientistOne [Meng et al., 2026] is an autonomous research system whose writer tags each sentence that holds a number with a line of its experiment log, and whose audit

reruns submitted code and compares the result with the reported score. `repro reproduce` differs from it in five ways. (1) *Who can use it.* The tags exist only for papers that ScientistOne itself writes, and its audit needs a benchmark's own evaluator; the `repro` manifest is written for any project, by a person or an agent, and lists the author's own commands. (2) *What the reader receives.* ScientistOne removes the tags from the final manuscript; the manifest, the pinned inputs and the log of reruns stay in the repository, so a reader repeats the check from a clone. (3) *How a check is decided.* Its audit has a language model read the reported score out of the paper, and two of its four checks are majority votes of a language model; every claim in the manifest is machine-checkable, in that the number the manuscript prints is compared, as a decimal at the precision printed, with the value at its declared address in the rerun output, and no language model takes part. (4) *What is fixed beforehand.* A regeneration runs only on inputs whose bytes match their pins, and a changed input is reported as `not re-run`; the ScientistOne paper describes no digest of an input or an output. (5) *What a result can be.* Its numeric check passes or fails, and its authors sort the misread values by hand; `unchecked`, `failed` and `not re-run` are outcomes of their own, so a check that could not be made is never counted with those that passed. ScientistTwo applies the same audit to its own output and reports every score verified on 49 target problems [Nam et al., 2026]. Its coding agent audits the repository against the manuscript, the rerun in the one audit report the paper reproduces covers the proposed method and not the baseline, and the paper states no tolerance for a score to count as verified.

Claims bound to files. SciTeX Clew links a manuscript claim to the session that produced its source file and traces backward from the claim through outputs, scripts and inputs to the raw data, verifying every hash [Watanabe, 2026]. PaperDelta traces a manuscript's statements to declared experiment evidence in exact decimal arithmetic and never runs the experiment scripts [PaperDelta contributors, 2026]. Both check a claim without a rerun. `repro reproduce` reruns the declared command, and the comparison of a printed value with its declared address is the last of three stages, so a rerun that did not happen and a value that could not be read are outcomes of their own.

Claims as records. Blueprints represent a project's claims, evidence, assumptions and definitions as typed graph components, and every structural rule over the graph is a warning [Ma, 2026]. Yin et al. [2026] propose a trace profile for autonomous scientific agents in which a claim is an ordinary record whose evidence bindings hold which field was read, which value was extracted and what comparison was made. `repro verify` makes that comparison against the file on disk, and a failed comparison stops the build under the strict policy.

Reproduction by an agent. SEER-Repro has a coding agent re-derive 2,190 atomic claims of 20 published cancer-registry papers from the registry data, and reproduces 77.4% of them [Liu et al., 2026]. A language model extracts each reproduced value from the agent's report, and deterministic code then applies the tolerance. The two designs start from opposite ends. In SEER-Repro a model extracts the claims from a published paper, the reference is the published number, and the analysis is rewritten by the agent; in the `repro` manifest the author declares each claim and its address, the reference is the project's own output, and the command that is rerun is the author's.

Audit by an agent. ReAgent audits research documents written by autonomous agents against their repositories [Shen et al., 2026]. Language-model agents write a claim contract for each claim, which records the claim type, the datasets, the metrics, the settings and the reported results, then inspect the code and rerun the experiments. On 96 agent-written repositories its audit decisions agree with manual annotation for 70.83%, and 28 of the 96 hold at least one static inconsistency. ReAgent asks whether the code implements the method the document describes, which `repro` does not check, and `repro` decides the comparison of a printed value with a file without a language model.

Formal verification in mathematics. Lean has a relatively small trusted kernel [de Moura and Ullrich, 2021]. A proof left unfinished holds the placeholder `sorry`, and `#print axioms` lists the axioms a theorem uses [Avigad et al., 2026]. A Lean blueprint relates the statements of a paper to their formal counterparts: a statement is marked when it is fully formalized, and the statements it uses form a dependency graph [Massot, 2020]. As in Lean, the check in `repro` is a small deterministic program whose outcome the author does not set, and what the record does not establish is listed beside what it does. An empirical claim has no proof, so the check is a comparison of a printed

value with a file, and `repro why` lists the files no command is recorded for and the choices nothing checks.

Checklists and badges. The machine learning reproducibility checklist asks an author to state what a paper includes [Pineau et al., 2020], and an ACM badge records that the main results were obtained in a later study by a team other than the authors [Association for Computing Machinery, 2020]. Neither specifies what is rerun or compares a printed number with an output. The `repro` manifest states what is rerun, as a list of commands and the input files each reads, and the log of `repro reproduce` records for every printed number whether the rerun output still holds it.

Verification in other fields. The philosophy of measurement separates an instrument's reading from the result inferred from the reading [Tal, 2017], and an extractor is an instrument in this sense: in Section 4.2, six extractors read the same bytes and returned six results. In formal verification, many early automated theorem provers performed proofs without recording them, so a result could not be checked afterwards [MacKenzie, 2001]; here every verdict is stored with its pinned artifacts.

L Related Work in Detail

Table 27 places Reproducible Science beside earlier work by the layer each system addresses. Section 6 treats claim-to-location binding, repository auditing, artifact-level verification, re-execution and entailment, and the paragraphs below take the rows it leaves out.

Table 27: Prior art by layer. The last column states what Reproducible Science does at that layer; a row marked *below* is expanded in this appendix, and a section reference names where the body or another appendix treats it. The claim-to-location row binds the unit Reproducible Science binds, a printed claim to a location inside a cited artifact. The other rows describe a whole artifact, a run, an event history or a semantic relation.

Layer	Prior art	Relation to Reproducible Science
entity, activity and agent provenance	PROV-O [Lebo et al., 2013], OPM [Moreau et al., 2011], PAV [Ciccarese et al., 2013]	defines no provenance ontology; exports an RO-Crate (below)
workflow structure	ProvONE [Cuevas-Vicenttín et al., 2016, Cao et al., 2016]	ordering as a constraint in the plan; below
event ordering	PROV constraints [Cheney et al., 2013]	consistency of a record with some ordering of its events; below
assertion-level publication	nanopublications [Groth et al., 2013, Kuhn et al., 2016]	an assertion published as its own object; below
signed attestation	in-toto [Torres-Arias et al., 2019], SLSA [Open Source Security Foundation, 2023], Sigstore [Newman et al., 2022]	no report is signed, and a frozen plan is dated by OpenTimestamps (Appendix G); below
packaging at rest	RO-Crate [Soiland-Reyes et al., 2022], Workflow Run RO-Crate [Leo et al., 2024]	export target: `repro export ro-crate`; below
input tracking	DVC [Iterative, Inc., 2020], AiiDA [Huber et al., 2020], tximeta [Love et al., 2020]	inputs pinned by digest, plus the address of each printed value; below
experiment tracking	Weights & Biases [Weights & Biases, 2026], MLflow [MLflow Project, 2026]	a manuscript sentence bound to a file address, not a metric to a run; below
execution capture	ReproZip [Chirigati et al., 2016], Binder [Jupyter et al., 2018]	which stored value backs which printed sentence; below
artifact-level verification	Code Ocean [Code Ocean, 2026], Whole Tale [Brinckman et al., 2019]	one outcome per assertion, not one per capsule or tale; Section 6
claim-to-evidence entailment	FEVER [Thorne et al., 2018], SciFact [Wadden et al., 2020], CliniFact [Zhang et al., 2025]	outcomes name a byte or decimal comparison, not a semantic label; Section 6
claim-to-work markup	ClaimReview [Schema.org, 2026, Google, 2026]	a claim addressed inside an artifact, not a whole work; below
claim-to-location binding	Paperclip [Generative Expert Labs, 2026], SciTeX Clew [Watanabe, 2026]	the same unit, compared deterministically against pinned bytes; Section 6
re-execution	DVC [Iterative, Inc., 2020], Snakemake [Köster and Rahmann, 2012], CODECHECK [Nüst and Eglen, 2021], ScientistOne [Meng et al., 2026], ReScience C [Rougier et al., 2017]	Appendix K
record linkage	Fellegi–Sunter [Fellegi and Sunter, 1969]	a third outcome for a decision not made; below
trace link recovery	information-retrieval traceability [Borg et al., 2013]	links declared when the sentence is written; below
write-time binding	Sweave [Leisch, 2002], showyourwork [Luger et al., 2021]	every printed value checked, generated or cited; below
repository auditing	adduce [Chudasama, 2026]	adduce runs `repro verify` as one of its rules; Section 6
agent trace graphs	LEDGER [Kim et al., 2026]	checks declared values; reconstructs no trace; below
epistemic dependence	Hardwig [1985], Wagenknecht [2014], Koskinen [2023], Ortmann [2025], Messeri and Crockett [2024]	one link made recomputable; below

Provenance vocabularies. PROV-O expresses the PROV Data Model in OWL2 as classes, properties and restrictions for representing and interchanging provenance information, and three classes form its basis: an entity, an activity that occurs over a period of time and acts upon entities, and an agent bearing responsibility for an activity [Lebo et al., 2013]. PAV extends PROV-O for the provenance, authoring and versioning of web resources, and distinguishes the contributors, authors and curators of content from the creators of its representations [Ciccarese et al., 2013]. Both vocabularies describe what happened to an artifact and who was responsible for it. In Reproducible Science, a manifest entry records where in an artifact a printed value sits, and the report records what a comparison at that address found.

Ordering in provenance standards. ProvONE separates prospective provenance, the workflow specification that produces a data artifact, from retrospective provenance, the steps that were executed [Cuevas-Vicenttín et al., 2016]. On the specification side a `Controller` states that one `Program`'s execution is controlled by another, so that in a synchronous dataflow model a program starts only once a preceding program terminates; on the trace side an `Execution` is associated with the `Program` it ran, which stands as its PROV plan [Cuevas-Vicenttín et al., 2016]. DataONE adopted the model to collect, store, index and browse the provenance of the data packages held on its member nodes [Cao et al., 2016]. OPM decorates a causality graph with time and does not derive causality from it, because time precedence does not imply causality: causal dependencies are asserted with edges, observed times serve to validate them, and all clocks are assumed to be synchronized [Moreau et al., 2011]. PROV's ordering constraints require a record to be consistent with a sensible ordering of its events, stated through precedence relations, and for that check the times attached to events are irrelevant [Cheney et al., 2013]. Table 28 sets the three standards beside the check Reproducible Science makes.

Table 28: How three provenance standards represent ordering, beside the check Reproducible Science makes. Each standard orders events within a single record. A confirmatory claim needs precedence between two records, a registration and a run, with no causal edge asserted between them.

Standard	How ordering appears	What it lets a reader conclude
OPM	observed times on a causality graph	that observed times agree with causal edges asserted separately
PROV constraints	precedence relations between events	that a record is consistent with some ordering of its events
ProvONE	control links in the workflow specification	which step must follow which in the plan
Reproducible Science	a run record read beside a pinned plan	that, as recorded, every run producing a confirmatory claim's evidence started after the plan's registration

The check Reproducible Science makes rests on precedence alone. It does not assert that a registration caused a run, and it reports a confirmatory claim `ordered`, `violated`, or `unchecked` with a reason, such as a missing run record or a plan changed since it was pinned. A plan rewritten after the results breaks its pin, and the claim's ordering reverts to `unchecked`, unless the freeze record is rewritten with it. The registration time is self-recorded, but an outside timestamp bounds when the plan existed (Section 7, Appendix G).

Signed attestation. In in-toto, whoever performs a step writes signed link metadata recording that the step took place, and verification checks that each step defined has enough signed links [Torres-Arias et al., 2019]. Verifying an artifact under SLSA means checking that a provenance statement's subject matches the artifact's digest and that the provenance matches expected values [Open Source Security Foundation, 2023]. Sigstore ties signatures to existing developer identities through OIDC, signs with ephemeral keys, and records signatures in artifact and identity logs [Newman et al., 2022]. Reproducible Science signs no report. A freeze sends the plan's digest, and nothing else, to the OpenTimestamps calendar servers (Appendix G), so the proof names no signer, which a plan frozen during double-blind review requires.

Packaging and content-addressed inputs. RO-Crate packages research artifacts with their metadata as Schema.org annotations in JSON-LD, in a structured archive of the items that contributed to a research outcome [Soiland-Reyes et al., 2022], and Workflow Run RO-Crate is a set of profiles that represent and package the provenance of a workflow run [Leo et al., 2024]. tximeta identifies the reference transcriptome of a quantification by a hashed checksum stored in the quantification output, and attaches annotation metadata when the file is imported [Love et al., 2020]. AiiDA records every calculation or workflow its engine runs in a provenance graph [Huber et al., 2020]. Reproducible Science also pins its inputs by digest, and adds the address of each value a manuscript prints from an output. `repro export ro-crate` writes our check records as an RO-Crate: each claim as a Schema.org `Claim`, each checked assertion as a `CheckAction`, and the latest rerun of a declared command as a `CreateAction`, in which case the crate also declares Process Run Crate. The fields for the address, the printed and stored values, and the outcome are our own additions, defined in a profile we publish; RO-Crate has no terms for them.

Experiment tracking. A Weights & Biases run is a record of a single computation, to which metrics, artifacts and other information are logged, and its artifacts track and version data as the inputs and outputs of runs [Weights & Biases, 2026]. MLflow Tracking logs parameters, code versions, metrics and output files, and each run records its metadata and artifacts [MLflow Project, 2026]. Both organize results by run. Reproducible Science ties a sentence of the paper to a pinned file, which can also point to a Weights & Biases run, an MLflow run, or another external record.

Execution capture. ReproZip traces one execution of an experiment and creates a self-contained package of all its required dependencies, which unpacks into a directory, a chroot, a virtual machine or a container [Chirigati et al., 2016]. Binder creates sharable, interactive environments in the cloud from existing repositories, capturing a repository and its environment in a container [Jupyter et al., 2018]. Capturing the environment shows that the code can run again, but not which stored number supports which printed sentence. Reproducible Science records that link as an assertion.

Replication by reimplementation. ReScience C is a peer-reviewed journal for computational research that encourages the explicit replication of published research through new, open-source implementations [Rougier et al., 2017]. A replication there writes the analysis again. `repro reproduce` reruns the authors' own declared commands and compares each printed number with the value at its declared address in the output (Appendix K).

ClaimReview. Schema.org defines ClaimReview as a fact-checking review of claims made or reported in a creative work, and reports it in use on 1K to 10K domains [Schema.org, 2026]. Google's fact-check structured data requires three of its properties: `claimReviewed`, a short summary of the claim; `reviewRating`, the assessment; and `url`, the page hosting the fact check [Google, 2026]. The claim therefore travels as text. A `Claim` records where it occurred through `appearance` and `firstAppearance`, and both take a `CreativeWork` [Schema.org, 2026], so the finest address the vocabulary offers is a whole work. Fact-checking projects use rating schemes that differ, each documented by its publisher, and search results show only the textual rating [Google, 2026]. Whether a review separates a claim found false from a claim a reviewer could not settle is therefore decided by each publisher, in its own words. Agreement between the markup and the page rests with the publisher: the guidelines state that "there must not be any mismatch between the structured data and page content" [Google, 2026], and the guidelines give no way to check it. `repro export ro-crate` writes each claim as a Schema.org `Claim` and records beside it the address inside the artifact and the outcome of the comparison made there.

Nanopublications. A nanopublication consists of an assertion, the provenance of the assertion, and publication information about the nanopublication itself, each an RDF graph, linked to the nanopublication through a head graph [Groth et al., 2013]. Nanopublications are stored in a decentralized server network and identified by trusty URIs, each carrying a hash computed on the RDF content it names, which makes a nanopublication and its whole reference tree verifiable and immutable, and servers only add entries and never update them [Kuhn et al., 2016]. A trusty URI plays the role a pin plays in Reproducible Science (Section 2). The units differ: a nanopublication publishes an assertion as a new citable object carrying its own provenance, and our evidence assertion adds nothing to the artifact it names, recording a location inside an artifact the manuscript already cites.

The evidence assertion as a contract. Meyer [1992] states a routine's obligations as a precondition that any call must satisfy, a postcondition the routine ensures in return, and a class invariant that constrains all routines of a class. Our evidence assertion carries all three. The precondition is that the manifest loads, so that every assertion's locator has the form its kind requires; a manifest that does not load reports nothing. The postcondition is that every assertion evaluated is reported with its three stages and exactly one of the five flattened labels of Table 3. The invariant holds across the four kinds of Table 2: a locator resolves to exactly one scalar or to none, two or more matches are reported as ambiguous rather than resolved to the first, and no backend searches a file for the printed number.

Record linkage and traceability. Fellegi and Sunter [1969] decide each record pair as a link, a non-link or a possible link, and a possible link requires expensive manual linkage. Comparing every pair of records would outstrip the capacity of even the largest and fastest computers, so they restrict the comparisons examined to a subspace of the pairs by blocking [Fellegi and Sunter, 1969]. Information-retrieval traceability recovers links between software artifacts in a similar way: candidate links are ranked by the similarity between artifacts, and an engineer examines each and confirms it or not [Borg et al., 2013]. Recovering a link after the fact produces some incorrect links [Borg et al., 2013]. In Reproducible Science, an evidence assertion is written with the sentence it binds, so no candidate set is ranked and no pair is searched for, and what remains is whether the value at the declared address matches.

Write-time binding. Sweave and showyourwork write each value into the document from the code that computes it, so the two match after every build [Leisch, 2002, Luger et al., 2021]. That holds only while the code still runs, and it often does not: Piccolo and Frampton [2016] trace irreproducible findings to how software is packaged, installed and executed, and to how analysis steps are documented. Reproducible Science checks each printed value against the stored result file, so the check works whether or not the code still runs.

Spreadsheets as a results format. Results are often published as Excel files: a screen of open-access genomics papers in PubMed Central from 2014 to 2020 found 32,841 with supplementary Excel files, and gene name errors appeared in 30.9% of the 11,117 that held gene lists, from gene names converted to dates, to floating-point numbers and to five-digit internal dates [Abeysooriya et al., 2021]. The table assertion of Reproducible Science reads such files directly, finding a cell by sheet, column and a selector that matches one row (Table 2).

Agent trace graphs. LEDGER is a tracing and review system that builds layered trace graphs over observed agent sessions [Kim et al., 2026]. Its authors observe that visibility alone leaves reviewers to reconstruct which actions, artifacts and validation steps matter for a conclusion, and its traces expose workflow decisions, artifact lineage, repair steps, validation coverage and claim-support paths for evidence-centered audit [Kim et al., 2026]. `repro verify` checks each declared value without a reviewer, but does not guarantee the right value was declared.

Epistemic dependence. Hardwig [1985] holds that a rational believer can never avoid some dependence on experts, and observes of a paper with 99 authors that many of them "will not even know how a given number in the article was arrived at." Wagenknecht [2014] separates two forms of that dependence within a research team: it is opaque when the dependent scientist lacks the expertise to carry out and assess a colleague's work, and translucent when she has it. Neither form is transparent, because in experimental science evidential contributions "can be entirely transparent only to those who have created or eye-witnessed them" [Wagenknecht, 2014]. For research that uses AI, Koskinen [2023] argues that no one can take informed responsibility for how an essentially opaque system reaches its output, so that its opacity "breaks the networks of trust assumed in social epistemology." Ortmann [2025] replies that relying on such a system requires reliability rather than trust, and cites language models paired with proof checkers, where the reliability of the final output does not depend on how often the model errs. Messeri and Crockett [2024] argue that scientists who trust AI tools to overcome their own cognitive limits become susceptible to illusions of understanding, and call for exposure to science and technology studies, social epistemology and philosophy of science. Reproducible Science checks reporting the way Ortmann's proof checker checks a proof: any reader can recompute the link between a printed value or quotation and its pinned artifact, while the extractor adds a dependence of its own (Appendix C).

M Integrations

Table 29 lists the systems Reproducible Science 0.7.0 reads from, with what each reader reads and writes and what it never does; each refusal there is enforced in code and checked by a test or a source line. Table 30 lists every other connection by where it runs, with its command and the release that first carried it. None of the readers is endorsed by or affiliated with the system it reads.

Table 29: Systems Reproducible Science 0.7.0 reads from. *Since* is the first release carrying the reader. A dash in the fourth column means no refusal backed by a test or a source line is recorded for that reader.

Integration	Command	What it reads and writes	What it never does	Since
CODECHECK [Nüst and Eglen, 2021]	`claims from-codecheck`	`codecheck.yml`, or a certificate ID against a pinned register; a draft manifest and open items	never writes a certificate field; binds no claim by itself; never overwrites	0.7.0
CORE-Bench [Siegel et al., 2024]	`claims from-corebench`	a capsule's results and its task file; a draft and open items with ranked candidates	never writes a candidate into the draft	0.7.0
ClawBio [Corpas, 2026]	`claims from-clawbio`	an output folder's checksums, report and tables; a draft and open items	—	0.7.0
PaperDelta [PaperDelta contributors, 2026]	`claims from-paperdelta`	`report.json` and its configuration; check records	refuses a configuration the report's digest does not match; imports no PaperDelta code	0.7.0
adduce [Chudasama, 2026]	rule `R-REPRO-001`	`repro.yaml`; a sidecar report and one finding with an item per checked assertion	never guesses among several manifests; never writes outside its sidecar	0.2.0
Outside records	`results annotate --kind`, e.g. `wandb:artifact`, `doi:dataset`, `zenodo:record`	the subject files; a ledger event linking them to the outside record	an annotation is never a seal	0.5.3

Table 30: Every other integration of Reproducible Science 0.7.0, grouped by where it runs, with its command and the first release that carried it. A skill is a set of instructions to the agent and enforces nothing.

Integration	Command	Since
Claude Code		
Plugins (marketplace, bundle)	`marketplace.json`, `reproducible-science`	0.1.0
Hook: frozen file	PreToolUse on Edit, Write	0.5.3
Hook: unfrozen plan	PreToolUse on Bash	0.4.1
Hooks: frozen plan changed, quotations, numbers	PostToolUse	0.3.0
Mod: frozen-plan guard	`tool.call` on Edit, Write	0.5.0
Mod: status	`/repro-status`, `/repro-verify`, `/repro-hide`, `/repro-show`	0.5.0
Mod: notes and project	`tool.call`; MCP tool `set_project`	0.5.0
Skills	`SKILL.md`: prereg, citations, results, repro, osf-registration, prereg-style	0.2.0
Open-items skill	`SKILL.md`: open-items	0.7.0
Codex		
Plugin	An experimental plugin for Codex, with the Claude Code plugin's five hooks and `/prereg-check`.	0.7.0
Any MCP client		
MCP server	`repro mcp`: runs the checks, pins quotations and proposes bindings, each proposal inferred until a person accepts it; it cannot freeze a plan, seal an input or run a declared command.	0.7.0

Integration	Command	Since
Git and CI		
Pre-commit	`repro-verify`, `repro-verify-strict`	0.2.0
CI workflows and checks	`repro ci init`; `repro ci prereg`, `results`, `claims`, `citations`, `reruns`	0.5.6
Re-run witness	`repro reproduce`	0.5.6
Pull-request bot	GitHub Actions: a comment per stale or broken claim with four numbered options; replies parsed without AI; options 3 and 4 written as `set_aside`; never edits the manuscript	0.7.0
Badges	`repro badges`, `repro ci endpoints`, `repro badges init`	0.6.0
Outputs		
JSON record	`repro export record`	0.5.6
SARIF	`repro verify --format sarif`; `repro export sarif`	0.2.0
RO-Crate [Soiland-Reyes et al., 2022]	`repro export ro-crate`	0.5.6
Registries and timestamps		
OSF token and draft	`prereg setup`; `prereg freeze --osf`	0.2.0
OSF registration	`prereg register`, `status`, `link`	0.5.0
OpenTimestamps [Todd, 2016], plans	`prereg freeze`, `prereg timestamp`	0.5.0
OpenTimestamps, ledger	`results timestamp`	0.5.0
Bibliographic services and sources		
Identifier lookup, registry audit	`citations resolve`; `citations audit`	0.2.0
Source download	`citations fetch`	0.5.0
Bibliography entry	`citations add`	0.4.0
Web page	`citations pin --url`	0.5.6
Paperclip [Generative Expert Labs, 2026]	`citations resolve --via paperclip`, `import-paperclip`	0.3.0
Manuscript formats		
Word (`.docx`) source	`citations verify`	0.5.0
Word (`.docx`) manuscript	`repro verify`	0.7.0
LaTeX (`.tex`)	read as text; this paper's quotations are checked against the text `scripts/detex.py` renders	
Markdown, text, reStructuredText (`.md`, `.txt`, `.rst`)	read as text by `citations verify` and `repro verify`	
Other repositories and open items		
Open items	`claims open-items`	0.7.0
Draft scaffold	`repro init`: given result files and a manuscript, writes `repro.draft.yaml` and `open-items.json` and binds nothing; the draft fails strict	0.7.0
Another repository	`repro audit`	0.5.0
Copied artifacts	`claims origins`	0.6.0

N Audit: Literature census of 43,462 abstracts

Description. A preregistered literature census of proteome-wide Mendelian randomization papers, in progress.[7] It counts how often a protein–outcome claim is repeated, and how often the repeats rest on overlapping data. The frozen plan is registered on the Open Science Framework (OSF) through `prereg register`, and the plan's log records the registration and each amendment. The repository carries 1 frozen plan with 194 log lines, 233 ledger events, 40 runs, 3,508 claim records and 23,939 quotations.

Example Finding. `results verify --files` reports one sealed test file as recorded with 2 different hashes: it was corrected in place and sealed again, against the study's rule that a fix is a new file. No command clears the report.

Table 31: Literature census: what each command found in the repository, and what it could not establish. Exit 0 is a pass, 1 a finding, and 2 nothing to read.

command	exit	what it found	what it could not establish
`prereg`	0	1 plan, unchanged against the digest recorded when it was frozen	whether the plan answers the question
`results`	1	chain intact at 233 events; 373 of 374 recorded files match their digest and 1 carries 2 recorded digests	which run a reported number came from, no number being in a manuscript yet
`citations`	1	13,989 of 13,989 pinned source sentences and 2,167 of 2,167 pinned sentences behind the dataset codes found in their stored sources; 7,783 quotations in the repository's own claims folder unchecked, their sources not being in the repository	whether a found sentence was read correctly
`repro`	0	5 of 5 assertions verified	any count of the census, the assertions covering the search counts the plan states

Application: Use during the study

The census is conducted with the toolkit. A census of this size, run by a language model and by coding agents in separate sessions, could not be audited or called confirmatory without the records of Table 32. Each extraction can be checked against its source sentence, and each rule is dated against what had been seen.

Table 32: The record kept at each stage of the census, and the command that keeps it. A review of this kind ordinarily reports one count per stage.

Stage	**Command**	**Record**
plan	`prereg`	one plan frozen before screening, and 194 dated log lines under it, each stating what had been seen when it was written
screening and extraction	`results`	90 seals of scripts and inputs and 40 recorded runs, over 43,462 abstracts screened and 14,053 extracted records
verification	`citations`	the source sentence of each of the 14,053 extracted records, checked against the stored abstract
dataset coding	`citations`	2,167 pinned sentences behind the codes for 872 dataset names
access to output	`results`	26 access records, each with what was shown and its level

Model extraction. A language model screens 43,462 abstracts and extracts 14,053 records from the 2,363 papers it admits. Each record carries the sentence it was drawn from, so `citations verify` decides for every record whether the model quoted the abstract. Of the 14,053 sentences,

[7] Registration: https://osf.io/zae8c. Repository: https://github.com/elliottower/proteome-mr-claim-audit, audited at commit `d833d00` (`62f181c` before the history rewrite described below).

13,989 resolve in the stored abstract. Of the 64 that do not, 27 occur twice in the abstract and 37 leave out one word of it. A found sentence establishes nothing about whether the model read it correctly.

Agent coding. Coding agents assign a dataset to each of 872 dataset names in 434 papers. An agent's code counts only when `citations pin` finds the sentence it rests on in the source and that sentence holds an identifier unique to the coded release. Across four passes of separate sessions, a script applies that rule to every code, and the coordinating agent asks each session for counts and file paths and never for values.

Access to output. Each access to output is a ledger record with a level. Of 26 accesses, 4 are at the level *metadata only*, 19 at *structure seen* and 3 at *outcomes seen*: a registered negative-control result, the text of three evidence sentences printed by accident during a count, and the opening lines of one batch of the second check, printed to learn its format. None shows a quantity either hypothesis concerns. The plan was frozen before all three, so each confirmatory number is still recorded as confirmatory, with the freeze it rests on named beside it.

Third-party text. The study first committed the abstracts and full texts it read, beside its own records. That text belongs to other publishers, so it was removed, which required rewriting the repository's history and changed every commit identifier, including the commit at which the plan was frozen. The public repository now lists each removed file with its path, its size and its SHA-256, and a script fetches it again. The ledger holds the same SHA-256, so a fetched abstract is checked against the bytes the model was given. The registration rests on three records that do not depend on the git history: the SHA-256 of the plan's text, written into the plan at the freeze; the copy registered on OSF; and the OpenTimestamps proof of that SHA-256 [Todd, 2016] (Appendix G). A registration pinned by a commit alone would not survive the same removal.